DBMS Mini project

Application: Student Record Management

1.Application interface:

Java code:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.Scanner;

public class StudentRecordManagement{

    static final String URL =
            "jdbc:mysql://localhost:3306/student_db";
    static final String USER = "root";
    static final String PASSWORD = "Root";

    static Scanner sc = new Scanner(System.in);

    public static Connection connect() throws SQLException {
        return DriverManager.getConnection(URL, USER, PASSWORD);
    }

    public static void addStudent() {
        String sql = "INSERT INTO students VALUES (?, ?, ?, ?, ?)";

        try (Connection con = connect();
             PreparedStatement ps = con.prepareStatement(sql)) {

            System.out.print("Enter student ID: ");
            int id = Integer.parseInt(sc.nextLine());
```

```
            System.out.print("Enter student name: ");
            String name = sc.nextLine();

            System.out.print("Enter email: ");
            String email = sc.nextLine();

            System.out.print("Enter course: ");
            String course = sc.nextLine();

            System.out.print("Enter semester: ");
            int semester = Integer.parseInt(sc.nextLine());

            if (id <= 0 || semester <= 0 || name.trim().isEmpty()) {
                System.out.println("Invalid input.");
                return;
            }

            ps.setInt(1, id);
            ps.setString(2, name);
            ps.setString(3, email);
            ps.setString(4, course);
            ps.setInt(5, semester);

            ps.executeUpdate();
            System.out.println("Student added successfully!");

        } catch (NumberFormatException e) {
            System.out.println("ID and semester must be numbers.");
        } catch (SQLException e) {
            System.out.println("Database error: " + e.getMessage());
        }
    }
```

```java
public static void viewStudents() {
    String sql = "SELECT * FROM students";

    try (Connection con = connect();
         PreparedStatement ps = con.prepareStatement(sql);
         ResultSet rs = ps.executeQuery()) {

        System.out.println("\n--- Student Records ---");

        boolean found = false;

        while (rs.next()) {
            found = true;
            System.out.println(
                "ID: " + rs.getInt("student_id")
                + " | Name: " + rs.getString("student_name")
                + " | Email: " + rs.getString("email")
                + " | Course: " + rs.getString("course")
                + " | Semester: " + rs.getInt("semester")
            );
        }

        if (!found) {
            System.out.println("No student records found.");
        }

    } catch (SQLException e) {
        System.out.println("Database error: " + e.getMessage());
    }
}
```

```java
public static void updateStudent() {
    String sql = "UPDATE students SET student_name=?, email=?, "
               + "course=?, semester=? WHERE student_id=?";

    try (Connection con = connect();
         PreparedStatement ps = con.prepareStatement(sql)) {

        System.out.print("Enter student ID to update: ");
        int id = Integer.parseInt(sc.nextLine());

        System.out.print("Enter new name: ");
        String name = sc.nextLine();

        System.out.print("Enter new email: ");
        String email = sc.nextLine();

        System.out.print("Enter new course: ");
        String course = sc.nextLine();

        System.out.print("Enter new semester: ");
        int semester = Integer.parseInt(sc.nextLine());

        if (id <= 0 || semester <= 0 || name.trim().isEmpty()) {
            System.out.println("Invalid input.");
            return;
        }

        ps.setString(1, name);
        ps.setString(2, email);
        ps.setString(3, course);
        ps.setInt(4, semester);
```

```
            ps.setInt(5, id);

            int rows = ps.executeUpdate();

            if (rows > 0) {
                System.out.println("Student updated successfully!");
            } else {
                System.out.println("Student ID not found.");
            }

        } catch (NumberFormatException e) {
            System.out.println("ID and semester must be numbers.");
        } catch (SQLException e) {
            System.out.println("Database error: " + e.getMessage());
        }
    }

    public static void deleteStudent() {
        String sql = "DELETE FROM students WHERE student_id=?";

        try (Connection con = connect();
             PreparedStatement ps = con.prepareStatement(sql)) {

            System.out.print("Enter student ID to delete: ");
            int id = Integer.parseInt(sc.nextLine());

            if (id <= 0) {
                System.out.println("Invalid student ID.");
                return;
            }
```

```
            ps.setInt(1, id);

            int rows = ps.executeUpdate();

            if (rows > 0) {
                System.out.println("Student deleted successfully!");
            } else {
                System.out.println("Student ID not found.");
            }

        } catch (NumberFormatException e) {
            System.out.println("Student ID must be a number.");
        } catch (SQLException e) {
            System.out.println("Database error: " + e.getMessage());
        }
    }

    public static void main(String[] args) {
        while (true) {
            System.out.println("\n===== STUDENT RECORD MANAGEMENT =====");
            System.out.println("1. Add Student");
            System.out.println("2. View Students");
            System.out.println("3. Update Student");
            System.out.println("4. Delete Student");
            System.out.println("5. Exit");
            System.out.print("Enter your choice: ");

            String choice = sc.nextLine();

            switch (choice) {
                case "1":
                    addStudent();
```

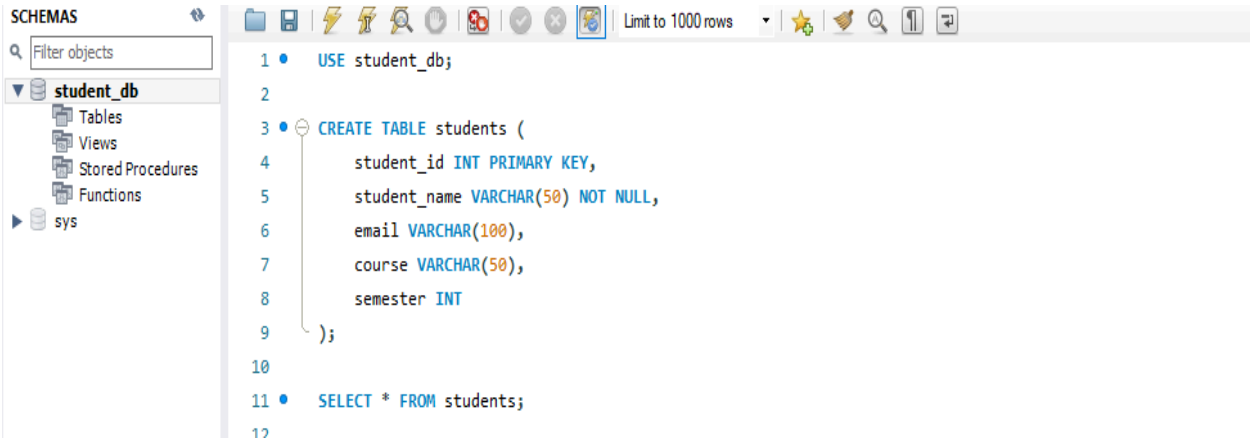
```
                    break;
                case "2":
                    viewStudents();
                    break;
                case "3":
                    updateStudent();
                    break;
                case "4":
                    deleteStudent();
                    break;
                case "5":
                    System.out.println("Thank you for using the application!");
                    sc.close();
                    return;
                default:
                    System.out.println("Invalid choice. Try again.");
            }
        }
    }
}
```

Output:

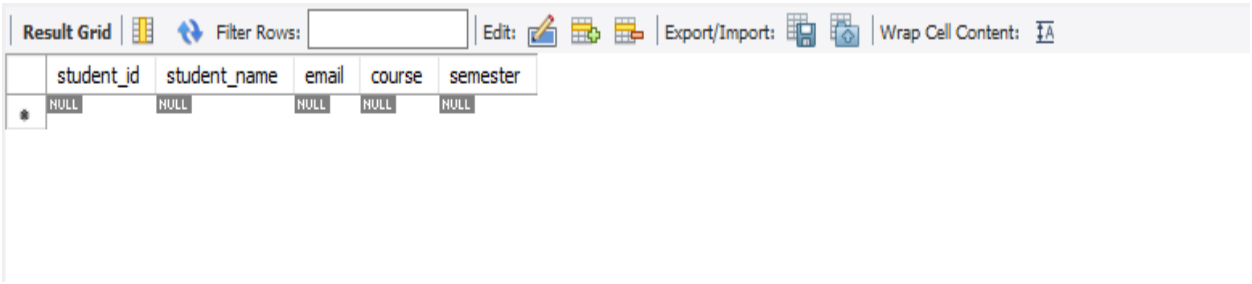
```
run:

===== STUDENT RECORD MANAGEMENT =====
1. Add Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter your choice:
```

2. MySQL database/table:



Output:



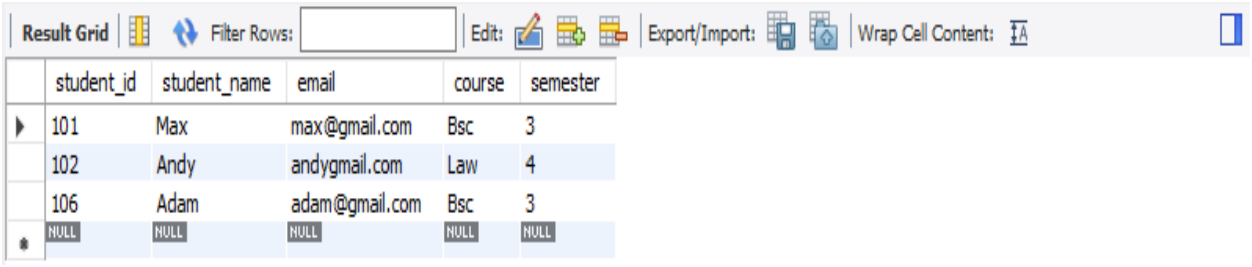
3. Data inserted through the application:

```
===== STUDENT RECORD MANAGEMENT =====
1. Add Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter your choice: 1
Enter student ID: 101
Enter student name: Max
Enter email: max@gmail.com
Enter course: Bsc
Enter semester: 3
Student added successfully!

===== STUDENT RECORD MANAGEMENT =====
1. Add Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter your choice: 1
Enter student ID: 102
Enter student name: Andy
Enter email: andygmail.com
Enter course: Law
Enter semester: 4
Student added successfully!

===== STUDENT RECORD MANAGEMENT =====
1. Add Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter your choice: 1
Enter student ID: 106
Enter student name: Adam
Enter email: adam@gmail.com
Enter course: Bsc
Enter semester: 3
Student added successfully!
```

Output: (In mysql)



student_id	student_name	email	course	semester
101	Max	max@gmail.com	Bsc	3
102	Andy	andygmail.com	Law	4
106	Adam	adam@gmail.com	Bsc	3
NULL	NULL	NULL	NULL	NULL

4. Retrieved/updated/deleted data:

```
===== STUDENT RECORD MANAGEMENT =====
1. Add Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter your choice: 2

--- Student Records ---
ID: 101 | Name: Max | Email: max@gmail.com | Course: Bsc | Semester: 3
ID: 102 | Name: Andy | Email: andygmail.com | Course: Law | Semester: 4
ID: 106 | Name: Adam | Email: adam@gmail.com | Course: Bsc | Semester: 3

===== STUDENT RECORD MANAGEMENT =====
1. Add Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter your choice: 3
Enter student ID to update: 106
Enter new name: Lily
Enter new email: lily@gmail.com
Enter new course: Design
Enter new semester: 4
Student updated successfully!


===== STUDENT RECORD MANAGEMENT =====
1. Add Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter your choice: 2

--- Student Records ---
ID: 101 | Name: Max | Email: max@gmail.com | Course: Bsc | Semester: 3
ID: 102 | Name: Andy | Email: andygmail.com | Course: Law | Semester: 4
ID: 106 | Name: Lily | Email: lily@gmail.com | Course: Design  | Semester: 4

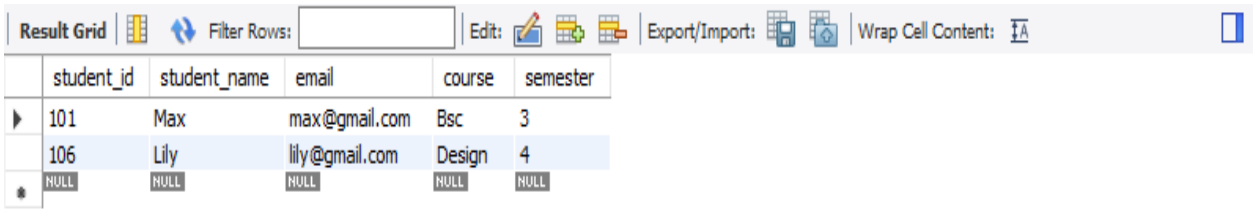
===== STUDENT RECORD MANAGEMENT =====
1. Add Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter your choice: 4
Enter student ID to delete: 102
Student deleted successfully!

===== STUDENT RECORD MANAGEMENT =====
1. Add Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter your choice: 2
```

```
--- Student Records ---
ID: 101 | Name: Max | Email: max@gmail.com | Course: Bsc | Semester: 3
ID: 106 | Name: Lily | Email: lily@gmail.com | Course: Design  | Semester: 4

===== STUDENT RECORD MANAGEMENT =====
1. Add Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter your choice: 5
Thank you for using the application!
```

Output: (In mysql)



student_id	student_name	email	course	semester
101	Max	max@gmail.com	Bsc	3
106	Lily	lily@gmail.com	Design	4
NULL	NULL	NULL	NULL	NULL